Predicting Calories in Fast Food Items from Macronutrients

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Motivation:

Fast food is becoming a big part of the modern diet due to its convenience and affordability, however it is also commonly associated with excessive caloric intake and health issues such as obesity, heart disease, and diabetes. Consumers of fast food are unaware about how nutritional components, such as fats, carbohydrates, sugar, and sodium contribute to the overall calories of fast food items, making it difficult to make informed decisions when it comes to dietary choices.

The goal of our project is to analyze nutritional information from fast food restaurants and their menu items and predict the calorie content of said items using the machine learning techniques we've learned in class. By modeling the relationship between nutritional information and calories, our project seeks to identify major drivers of caloric density in fast food menu items. For us to be able to understand the relationship between the two is not only beneficial for consumers, but also health researchers and food policymakers in making decisions about health risks associated with fast food.

Problem Translation:

We translated the problem of understanding caloric content in fast food items into a supervised machine learning regression problem. Each object in the dataset represents a fast food menu item along with its nutritional attributes. The goal here is to use these attributes as input features to predict the caloric value associated with each respective food item.

In this project, the target variable is the number of calories of food items, while the input features are the nutritional information, i.e. total fat, carbohydrates, sugars, proteins, sodium, etc.

Features that were irrelevant to the data, such as the company, item type, and weight watchers points associated with it were dropped. Calories from fat were dropped due to the large amount of missing values associated with the feature, as well as the calories from fat being a direct function of both Calories and Fat.

Because the target variable is continuous, regression models as taught in the course are appropriate to be used for this task. We will then train and compare the models with one another to evaluate how different models capture the relationship between nutritional attributes and calories.

Data Description:

Note: Prior to doing sections of the EDA, a small amount of data cleaning was needed to ensure proper formatting issues that the dataset had. Things such as the numeric columns being stored as strings and missing values in the dataset. In order to make any meaningful visualizations, this cleaning was mandatory.

This dataset contains information from Mcdonald's, Burger King, Wendy's, Kentucky Fried Chicken, Taco Bell, and Pizza Hut. Initially containing 1148 pieces of data with 14 variables in each observation. The continuous target variable Calories are the units of energy that the body uses for daily functioning. The nutrient features are all different macronutrients that are in each food item and affect the health and nutritional value of the food.

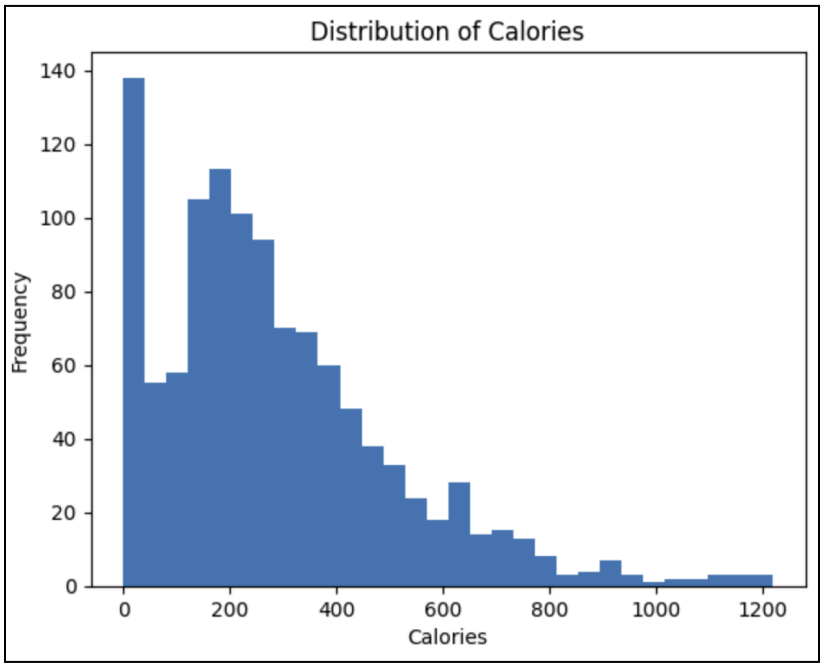
Mean Values of 5 Relevant Food Categories

Calories	Total Fat (g)	Carbs (g)	Fiber (g)	Protein (g)
287	11.7	38.9	1.45	9.43

The chart above contains the mean amount of 5 of the features in the dataset. However, it is important to note that with such a large amount of items, the ranges of each feature are very

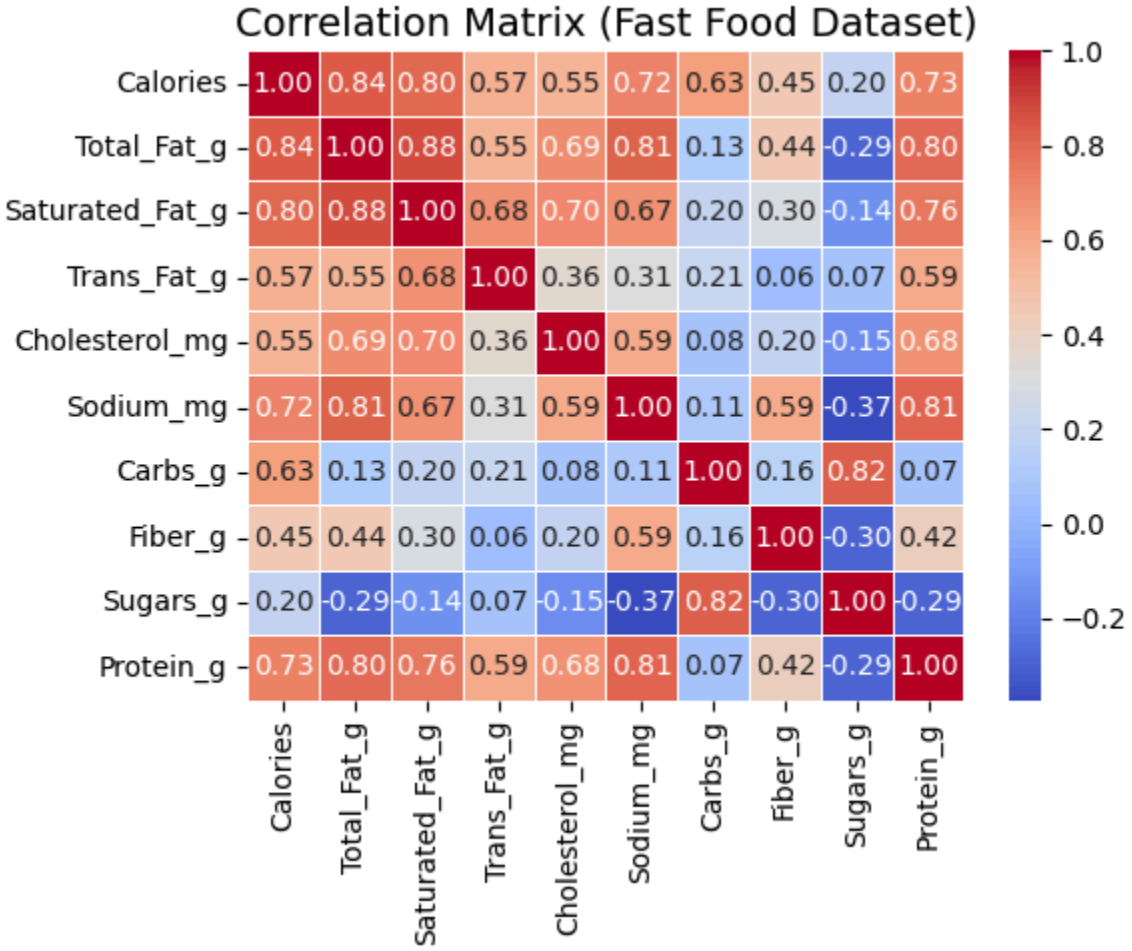
high, for example, the calories range from 0 calorie diet sodas to high calorie hamburgers. This information was found using df.describe().

	Calories	Total_Fat_g	Saturated_Fat_g	Trans_Fat_g	Cholesterol_mg	Sodium_mg	Carbs_g	Fiber_g	Sugars_g	Protein_g
count	1133.000000	1079.000000	1078.000000	1079.000000	1119.000000	1132.000000	1078.000000	1064.000000	1118.000000	1079.000000
mean	287.356575	11.682576	4.066790	0.140871	40.647900	427.722615	38.946197	1.454887	24.067979	9.434662
std	222.135591	13.957331	5.027242	0.489773	68.058662	496.701818	31.654807	2.389581	32.369327	11.049859
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	140.000000	0.000000	0.000000	0.000000	0.000000	70.000000	17.000000	0.000000	2.000000	0.000000
50%	240.000000	8.000000	3.000000	0.000000	20.000000	190.000000	34.000000	0.000000	8.000000	7.000000
75%	390.000000	18.000000	6.000000	0.000000	50.000000	680.000000	52.000000	2.000000	40.000000	14.000000
max	1220.000000	98.000000	33.000000	4.500000	575.000000	2890.000000	270.000000	31.000000	264.000000	71.000000

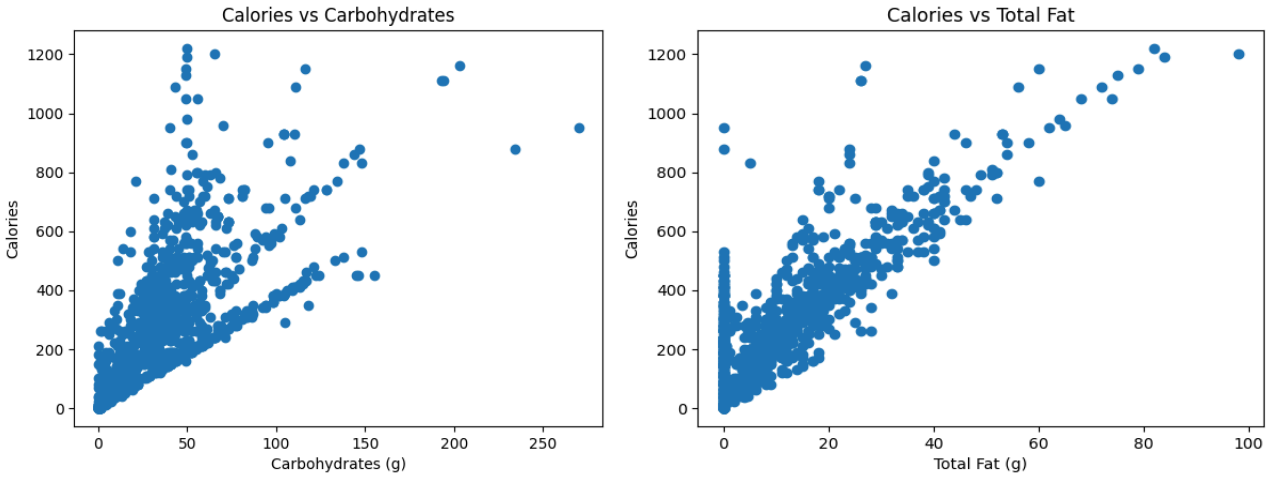


As seen in the chart above, the range of caloric items greatly varies. The information is right skewed with a median value of 240 lower than the mean amount of calories 287. This is because fast food chains often have similar offerings in drinks that share the same nutritional facts and while Sodas are unhealthy, they are lower in calories due to their nature as drinks.

Calories have a positive correlation with all of the macronutrients featured in the dataset.



The red boxes represent the variables that contain the strongest correlations, with Calories having the most consistent correlation with all of the other nutritional factors because each of the macronutrients are a part of the overall Calories, while the other macronutrients correlating with each other are due to coincidence.



The graphs above display the positive correlation that calories has with two macronutrients on a scatterplot. As total fat increases, the amount of calories increases and the same trend is seen in carbohydrates vs calories. This corroborates the above correlation matrix, showing the positive relationship between it and the target variables.



This boxplot represents the calorie distribution, dropping all items that contain 0 calories. Even with the 0 calorie items removed, 75% of fast food items contain less than 400 calories as the third quartile boundary on the box ends below the 400 point. The data set contains outliers as the data nears 800 calories.

Data Preparation:

The dataset has the variables Calories from fat, Weight Watchers Points, Company, and Item. Our regressions are only to find the relationship between macronutrients and calories, so we can safely remove these variables to narrow our data down to relevant to our topic at hand. We also remove all the data with a calorie total equal to 0 because the only food items with

calorie totals that low are diet sodas and water which can be excluded from this data. To split the data, the X variable was set to the macronutrients in the data table as the X variable is the inputs. The Y variable is the expected outcome which is why Calories was separated from the rest of the data and selected to be the Y variable.

Selected Models:

1. Linear Regression: Used to establish a very simple relationship between calories and nutritional contents
2. Non-linear/Polynomial Regression: Used to capture any potential non-linear relationships in the data while still being easily interpretable.
3. K-Nearest Neighbors Regression: Used to capture local patterns in the data without the assumption of any linear relationships.
4. Support Vector Regression: Used to model any complex, non-linear relationships.
5. Decision Tree Regression: Used to model non-linear feature relationships while maintaining interpretability.
6. Random Forest Regression: Used to reduce overfitting and improve prediction accuracy

Evaluation Metrics:

- Root Mean Squared Error (RMSE): This metric was used to measure model error in the same units as the target variable (i.e. calories). This standardization makes the results easy to interpret given our real-world context. RMSE also represents large errors well and penalizes it, which is good in our case because we don't want large calorie mispredictions.
- R^2: This metric was used to evaluate how well (or not well) each model represents the variance in caloric values. It directly indicates how well the model fits the data therefore

it provides a good straightforward comparison that can be used to compare models with one another

Hyperparameter Optimization (HPO):

There were two models that required HPO, these being the Decision Tree Regressor and the Random Forest Regressor. They needed tuning because unlike the other models, these models are very sensitive to hyperparameters.

1. Decision Tree Regression ⇒ The hyperparameter tuned was the minimum number of samples required in a leaf node. Smaller values generally risk overfitting, which is why we optimized this. Like we did in class, the procedure was taking candidate values and evaluating the validation R^2 score, with the highest score being selected for the model.
2. Random Forest Regression ⇒ The hyperparameter tuned was the number of trees in the forest (n_estimators). The reason why it was tuned is because generally the higher number of trees, the higher the performance, but at a certain point there are diminishing returns, so we want the lowest number of trees possible with the highest performance possible. We followed the same procedure like we did in class and defined candidate values and measured their performance with R^2, the value for n_estimators with the highest score was ultimately chosen.

Model Evaluation:

Training Performances for All Models

	Model	RMSE	R2
0	Linear Regression	12.305043	0.996785
1	Polynomial Regression	10.302455	0.997746
2	KNN Regression	38.084280	0.969202
3	SVR	161.796496	0.444136
4	Decision Tree	12.829704	0.996505
5	Random Forest	12.829704	0.996505

As shown by the training results, linear regression and polynomial regression both had outstanding results with very high R^2 values alongside low RMSE values. Decision Tree and Random Forest also had pretty good results, but just slightly lower performance than linear regression and polynomial regression in both metrics, it is also to note that their metrics are equal to each other. KNN had a decent performance for R^2, but it suffers in RMSE with it being around 38. We also see that the SVR model suffered greatly with lackluster performance in both metrics.

Validation Performances for All Models

	Model	RMSE	R2
0	Linear Regression	13.793413	0.996437
1	Polynomial Regression	11.622651	0.997470
2	KNN Regression	58.847536	0.935146
3	SVR	177.330464	0.411091
4	Decision Tree	42.052549	0.966882
5	Random Forest	42.052549	0.966882

Now looking at the validation results, polynomial regression showed the greatest performance with the lowest RMSE and the highest R^2 out of all models. Linear regression was a close second with similar results. Again, Decision Tree and Random Forest had an acceptable performance but

not to the likes of the previous two aforementioned. KNN has relatively mediocre performance and SVR has abysmal performance for these results

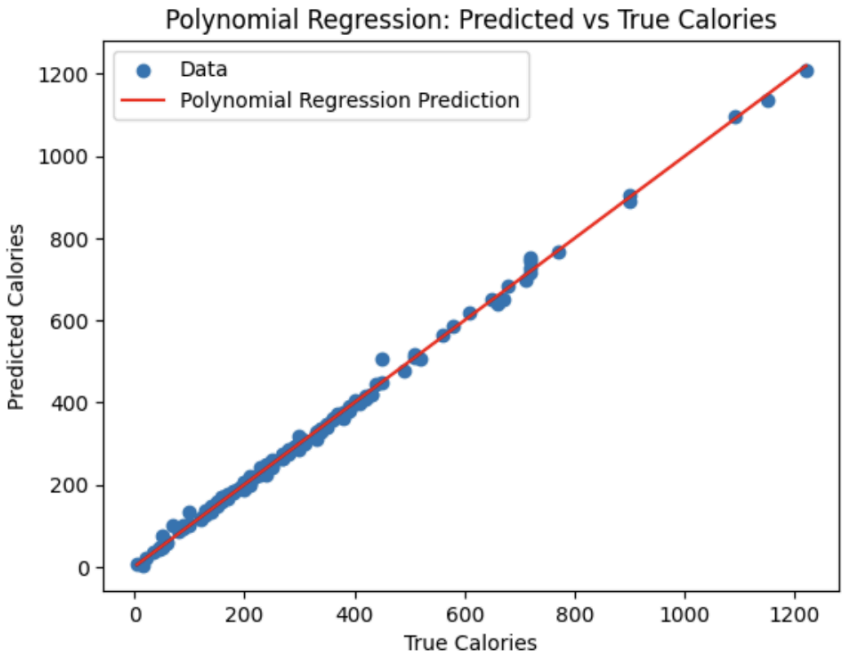
Test Performances for Top 3 Models

	Model	RMSE	R2
0	Polynomial Regression	11.622651	0.997470
1	Linear Regression	13.793413	0.996437
2	Decision Tree	42.052549	0.966882

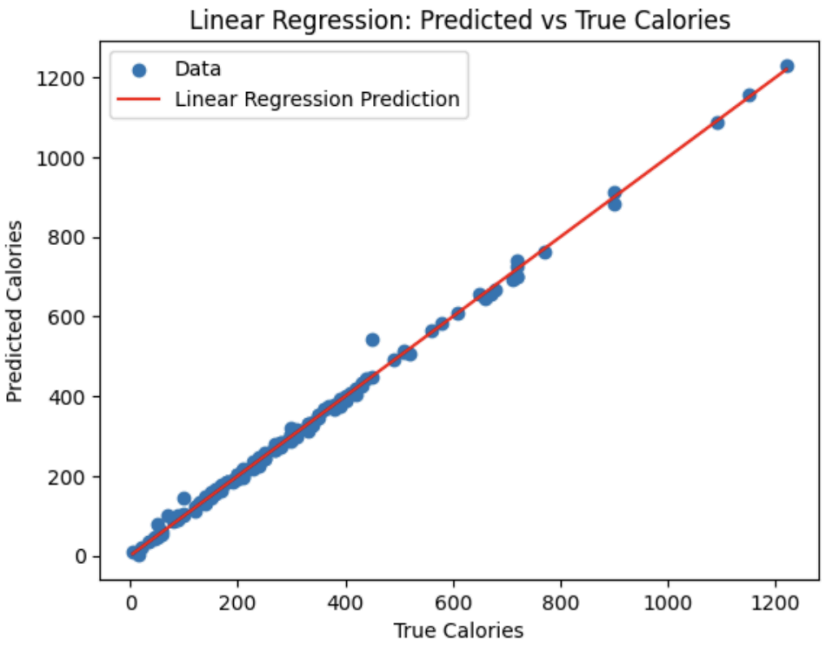
The top 3 models were evaluated based on their test performances. In the test performance results we notice the pattern remains intact, with polynomial regression being at the top, linear regression following slightly behind, and Decision Tree coming in third.

Overall, Polynomial Regression is the best model for this dataset on both metrics of RMSE and R^2 when put up against Linear Regression, K-Nearest Neighbors, SVR, Decision Tree, and Random Forest models.

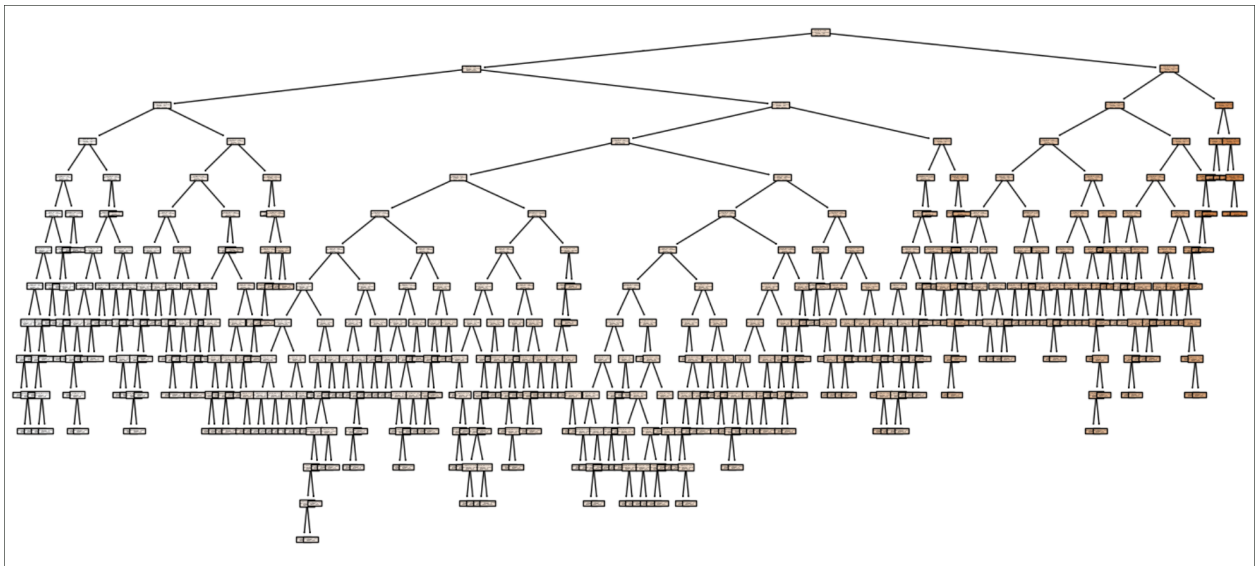
Model Predictions:



This visualization for predicted vs true calories for the polynomial regression model shows an extremely strong alignment between the two variables. What this means is the model successfully captures the nonlinear relationships within the dataset between nutritional contents and the number of calories without sacrificing low error. There is more deviation as the calorie count goes up however, which suggests that the model has more difficulty predicting more extreme values of calories.



The model for linear regression also shows a satisfactory fit. The predictions closely follow the true calories across the data. The almost perfect linear model suggests that there is a strong positive correlation between calories and nutritional attributes. We can see that like the polynomial model, this model also seems to deviate more at more extreme values. We can also see that the predictions are slightly more variable overall than the polynomial model.



Pictured above is the visualization of the decision tree model for this dataset. Although the nodes themselves can't be seen directly, the structure of the tree still tells us a lot about how the model makes predictions; it recursively splits data based on its features (in this case nutritional attributes). We can also see that the depth and complexity of the tree is pretty daunting which could indicate that it is overfitting. This would explain why it had worse performance compared to the polynomial regression and linear regression models.